

Contoso Electronics

Equipment Reference & Maintenance Guide

Wave Soldering Machine

Contoso FlowSolder WS-600

Asset Identification

Field	Detail
Asset ID	CE-WAV-2600
Category	Through-Hole Line
Manufacturer	Contoso Assembly Tools
Location	Through-Hole Line 2 - Position 4
Installation Date	2020-06-05
Warranty Expiry	2025-06-05
Support Contact	through.hole.support@contoso.com ext. 4540

Overview

The Contoso FlowSolder WS-600 is a wave soldering machine for through-hole and mixed-technology assemblies. It applies flux, preheats the board, and passes it over a molten solder wave to form reliable solder joints at production volumes.

Technical Specifications

Parameter	Value
Solder Pot Capacity	250 kg
Max Board Width	400 mm
Conveyor Speed	0.5 - 2.0 m/min
Preheat Zones	3 (IR + convection)
Flux System	Spray fluxer
Solder Alloy	SAC305 lead-free
Power	3-phase 400 VAC, 30 kW

Preventive Maintenance Schedule

Interval	Task
Daily	Check solder level and dross; verify flux spray pattern.
Weekly	Skim dross from solder pot; clean fingers/pallets.
Monthly	Analyze solder alloy composition; clean flux nozzles.

Quarterly	Inspect pump impeller and preheater elements.
-----------	---

Safety Guidelines

- Molten solder pot exceeds 260 °C - wear face shield and heat gloves when skimming.
- Flux fumes are hazardous - ensure extraction is active.
- Lead-free alloy is hot and heavy - use proper dross tools.
- Lockout/tagout before servicing the conveyor or pump.

Common Issues & Troubleshooting

Symptom	Recommended Action
Solder skips/voids	Check wave height, flux coverage, and preheat profile.
Excessive dross	Reduce wave turbulence; verify nitrogen (if equipped).
Bridging	Adjust conveyor speed and exit angle; check board design.
Poor hole fill	Increase preheat; verify flux activity and contact time.

Contoso Electronics internal maintenance documentation. For service requests, contact through.hole.support@contoso.com | ext. 4540 or submit a work order through the maintenance portal.